

ENG-654 · LECTURE 07

Workspace Path Planning

Follow a supplied workspace path or trajectory with a feasible joint motion

The desired motion is supplied by a precomputed trajectory or a real-time policy. We study planning methods for noncuspidal and cuspidal robots.

We study workspace path planning with a given path or trajectory

Assume that a desired workspace path or trajectory has been given to the robot to follow. It can come from either source below.

1 · Precomputed trajectories

The desired motion is calculated in advance and supplied as a sequence of workspace targets, with timing when a trajectory is specified.

The complete command is available for analysis before execution.

2 · Real-time policies

A policy uses current observations to provide the next desired workspace pose or motion command while the robot moves.

Only the current command or a short future horizon may be available.

A path specifies where to move; a trajectory also specifies when. Our task is to find a continuous joint motion that follows the supplied command while checking singularities and joint limits.

Find feasible joint motions for the supplied workspace command

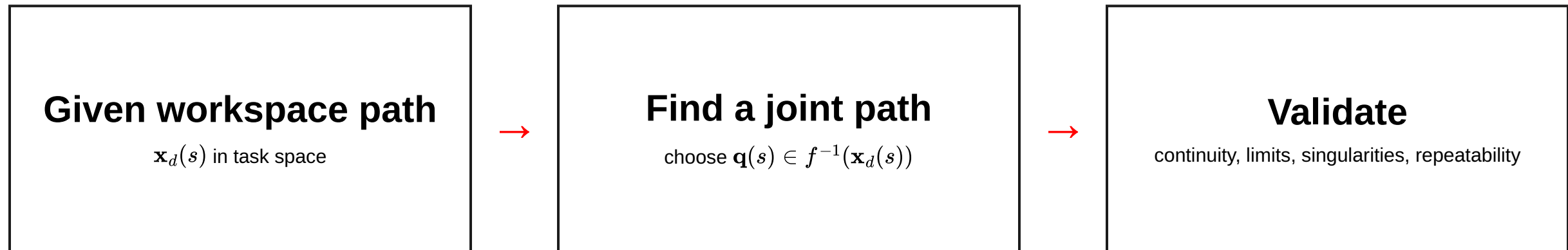
01 **Numerical and analytical planning**
Local continuation, complete inverse solutions, and connectivity.

02 **Noncuspidal robots · ABB IRB 4600**
Hybrid planning or analytical shoulder–elbow–wrist classification.

03 **Cuspidal 3R robots · custom_3R.urdf**
One task-space curve can produce different joint-space outcomes.

04 **Cuspidal 6R robots · ABB GoFa CRB 15000**
Inverse-solution maps, joint limits, and connections between configurations.

A workspace curve is not yet a robot trajectory



“Lift” means the continuous joint trajectory $\mathbf{q}(s)$ that realizes the chosen task-space curve: $f(\mathbf{q}(s)) = \mathbf{x}_d(s)$. It is a mathematical mapping, not an upward motion. The same Cartesian curve can have several such joint paths, each with a different outcome.

Build a joint path using local continuation or complete inverse solutions

Numerical continuation

$${}^0\mathcal{V}_t = \begin{bmatrix} {}^0\dot{\mathbf{p}}_t \\ {}^0\boldsymbol{\omega}_t \end{bmatrix} = {}^0J_t(\mathbf{q})\dot{\mathbf{q}}$$

Solve the velocity equation near the current configuration. Predict the next joint state, correct its FK error, and validate that step.

One initial IK selects one local trajectory.

Analytical or algebraic IK

$$\mathcal{Q}_k = f^{-1}({}^0T_{t,d}(t_k))$$

Enumerate every real solution at each target pose using closed-form expressions or an algebraic root solver. Connect adjacent solutions continuously.

Enumeration reveals candidates; connectivity and constraints determine feasible paths.

For a 6R robot, “ $\dot{\mathbf{x}} = J\dot{\mathbf{q}}$ ” uses the geometric velocity $[\dot{\mathbf{p}}; \boldsymbol{\omega}]$. An Euler-angle derivative needs its representation-dependent angular mapping.

Enumeration gives global candidates; continuation gives local tracking

	Jacobian / numerical	Analytical branch evaluation
Information used	Local derivative at the current $\mathbf{q}$	Closed-form or algebraic inverse structure for the modeled geometry
Initialization	A seed configuration	Desired pose and robot parameters; labels organize the returned roots
Output	One nearby solution, if convergence succeeds	All evaluated real roots, then a selected branch
Singularity signal	Conditioning degrades during iteration	Branch discriminant or separator vanishes
Scope	LOCAL TRACKING	GLOBAL ROOT ENUMERATION, NOT GLOBAL PATH CERTIFICATION

Local safeguards detect failure; they cannot select a globally viable start

Step size

Large Cartesian steps invalidate the local linear model and can jump to another basin.

Near singularity

Stop when the full Jacobian becomes singular or ill-conditioned. Damping can moderate a numerical step, but cannot restore a missing task direction.

Joint limits

A regular branch may still hit a physical joint stop. Global comparison can select a different initial IK that completes the path; it does not remove the stop.

$\|\dot{\mathbf{q}}\|$ can grow while $\|\dot{\mathbf{x}}\|$ stays modest as $\sigma_{\min}(J) \rightarrow 0$.

For this noncuspidal ABB, classify the solutions by shoulder, elbow, and wrist

Shoulder · S±

Choose the shoulder branch that places the wrist centre on the required side of the first axis.

Elbow · E±

Choose one of the two arm solutions for that shoulder branch.

Wrist · W±

Choose the wrist solution that realizes the desired tool orientation.

$$2 \times 2 \times 2 = 8 \text{ generic mathematical IK solutions}$$

Away from singularities, the S/E/W classification distinguishes the ABB IRB 4600 branches. Hybrid planning enumerates the starts once and continues them numerically. Fully analytical planning evaluates the same branch labels at every sample. Both methods must check joint limits and singularities.

An xy rectangle crosses regions with different numbers of positional IKs

The tool follows a 0.85 m × 0.55 m rectangle at $z = 1.10$ m with fixed downward orientation. The arm has four IKs in the inner region and two in the outer region; each regular arm solution supplies two wrist solutions.

Observe: compare the desired and achieved tool points. The shaded count is for the wrist-centre arm solutions (q_1, q_2, q_3); watch it change from four to two across the boundary.

ABB IRB 4600 · fixed tool orientation · tool path

STL opacity 100%

Tool xy path · z = 1.10 m

Position IKs: 4 2 0

— desired

— achieved

x [m]

Play path

Reset

S+ E+ W+

Duration T 12 s

Path progress

Start IK	Local continuation
S+ E+ W+	Loop complete
S+ E+ W-	Loop complete
S+ E- W+	Rejected: q3 > 75°
S+ E- W-	Rejected: q3 > 75°
S- E+ W+	Rejected: q2 < -90°
S- E+ W-	Rejected: q2 < -90°
S- E- W+	Stopped: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary
S- E- W-	Stopped: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary

s = 1.000 · S+ E+ W+ · hybrid predictor + Newton

4 position IKs · 8 full-pose IKs at the desired point

q_{1...q₆} [°]: 12.1, 1.3, 22.6, 0.0, 66.1, -167.9

|det ⁰J₆| = 1.7768 m³ · FK error 8.3e-11

q̇ [°/s]: 0.0, 0.0, 0.0, 0.0, 0.0, 0.0

T=12s · peak speed / limit = 0.26 ✓ (sampled whole path)

Analytical IK initializes each S/E/W branch; J q̇ = V predicts motion and Newton corrects it. A separate smooth clock stops at each rectangle corner. Increasing T can fix a speed excess; it cannot restore a missing IK branch.

Start with all analytical IKs, then follow each with the Jacobian

$$\begin{bmatrix} {}^0\mathbf{v}_6 \\ {}^0\boldsymbol{\omega}_6 \end{bmatrix} = {}^0J_6(\mathbf{q})\dot{\mathbf{q}}$$

$${}^0J_6(\mathbf{q}_k)\Delta\mathbf{q} = \begin{bmatrix} {}^0\mathbf{p}_{6,d} - {}^0\mathbf{p}_6 \\ \text{Log}(R_d R^\top)^\vee \end{bmatrix}$$

$$|\det {}^0J_6| < \varepsilon \implies \text{stop this continuation}$$

Evaluate the twist at the tool. For this robot, the tool origin is O_6 . With ${}^0\mathbf{r}_{56} = {}^0\mathbf{p}_6 - {}^0\mathbf{p}_5$, shift the wrist

Jacobian to the tool: ${}^0J_6 = \begin{bmatrix} I & -[{}^0\mathbf{r}_{56}]_\times \\ 0 & I \end{bmatrix} {}^0J_5$. Here

$[\mathbf{r}]_\times \mathbf{a} = \mathbf{r} \times \mathbf{a}$, and $\det {}^0J_6 = \det {}^0J_5$.

Here $R = {}^0R_{\text{tool}}$ and $R_d = {}^0R_{\text{tool},d}$ are the current and desired tool orientations. $\text{Log}(R_d R^\top)^\vee$ is their axis-angle error vector in the world frame.

- 1** Solve all IKs analytically at the first pose. Classify S, E, W and retain starts inside the joint limits.
- 2** For each start, compute ${}^0J_6\dot{\mathbf{q}} = [{}^0\dot{\mathbf{p}}_{6,d}; {}^0\boldsymbol{\omega}_d]$, then predict $\mathbf{q}_{k+1} \approx \mathbf{q}_k + \Delta t \dot{\mathbf{q}}_k$.
- 3** Correct the predicted pose by Newton iterations using the displayed position and orientation error.
- 4** Check singularities, position limits, and joint-speed demands along the whole path. Slow down for a speed limit; choose another branch when a joint stop blocks the path.

Four legal starts: two finish; two reach a positional singularity

Analytical IK supplies the starting solutions; each numerical predictor and correction uses the current Jacobian. Compare every starting IK on the same rectangle: successful local convergence at the start does not ensure completion.

ABB IRB 4600 · fixed tool orientation · tool path

STL opacity

100%

Tool xy path · z = 1.10 m

Position IKs:

□ 4

■ 2

■ 0

— desired

— achieved

1

0.75

0.5

0.25

0

1.25

1.5

1.75

2

2.25

2.5

y [m]

start

x [m]

Play path

Reset

S- E- W+ ▾

Duration T 12 s

Path progress

Start IK	Local continuation
S+ E+ W+	Tracking
S+ E+ W-	Tracking
S+ E- W+	Rejected: q3 > 75°
S+ E- W-	Rejected: q3 > 75°
S- E+ W+	Rejected: q2 < -90°
S- E+ W-	Rejected: q2 < -90°
S- E- W+	Stopped: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary
S- E- W-	Stopped: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary

s = 0.197 · S- E- W+ · hybrid predictor + Newton

2 position IKs · 4 full-pose IKs at the desired point

q1...q6 [°]: -171.7, -70.1, -85.5, 180.0, 114.5, -171.7

|det °J6| = 0.1545 m³ · FK error 2.8e-12

q̇ [°/s]: -1.7, -367.4, 675.2, -0.0, 307.8, -1.7

T=12s · peak speed / limit = 3.86 – increase T (sampled admissible prefix)

This branch reaches a positional fold. The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary. The robot holds its last admissible configuration. Choose S+ E+ to complete the same command.

Compare all ABB solution branches along the entire rectangle

Analytical IK enumerates the full-pose roots at each path sample. Continuous branch tracking and joint-limit checks identify the initial configurations that can finish.

ABB IRB 4600 · fixed tool orientation · tool path

STL opacity 100%

Tool xy path · z = 1.10 m

Position IKs: 4 2 0
— desired — achieved

x [m]

Play path Reset S+ E+ W+ ▾

Duration T 12 s

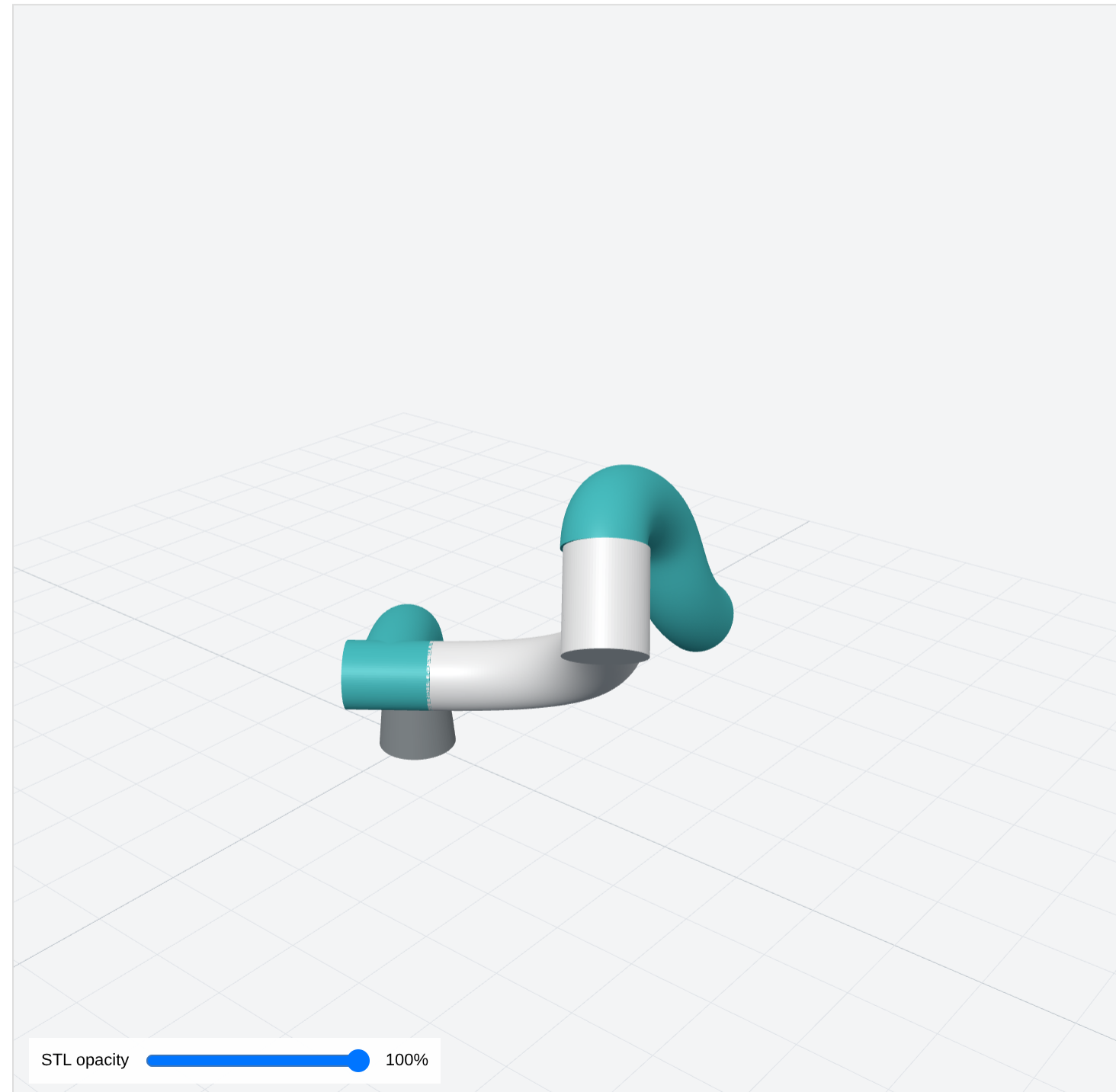
Path progress

Start IK	Whole-path analysis
S+ E+ W+	Completes loop
S+ E+ W-	Completes loop
S+ E- W+	Rejected: q3 > 75°
S+ E- W-	Rejected: q3 > 75°
S- E+ W+	Rejected: q2 < -90°
S- E+ W-	Rejected: q2 < -90°
S- E- W+	Stops s=0.197: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary
S- E- W-	Stops s=0.197: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary

s = 1.000 · S+ E+ W+ · analytical IK
4 position IKs · 8 full-pose IKs at the desired point
q1...q6 [°]: 12.1, 1.3, 22.6, -0.0, 66.1, -167.9
|det °J6| = 1.7768 m³ · FK error 3.6e-16
q̇ [°/s]: 0.0, 0.0, 0.0, 0.0, 0.0, 0.0
T=12s · peak speed / limit = 0.26 ✓ (sampled whole path)

8 mathematical starts → 4 admissible starts → 2 complete sampled joint paths. All IK roots are recomputed at every pose; limits and continuation decide which starts work.

Use custom_3R.urdf to connect workspace geometry and joint-space paths



The actual course robot

The URDF supplies the joint axes, offsets, and CAD geometry. The inverse solver and singularity curves use those same parameters.

Workspace event

The verified example starts inside a four-IK region and encircles the cusp near $(\rho, z) = (2.73, 2.93)$ m. A critical-value crossing can make some inverse solutions singular while others remain regular.

Planning consequence

For the course robot, different lifts of the same loop can close, terminate, or change IK solution without becoming singular.

Edit one path; carry its exact vertices into the following slides

Move or reshape the default loop, or draw vertices in the URDF (ρ, z) slice. Red curves are critical values; $\times$ marks the selected cusp. The start is marked separately.

Workflow: finalize one shared path. Every following 3R panel recalculates its joint trajectories. If your path has no example of a given type, that panel says so.

One shared task-space path · $y = 0$

-

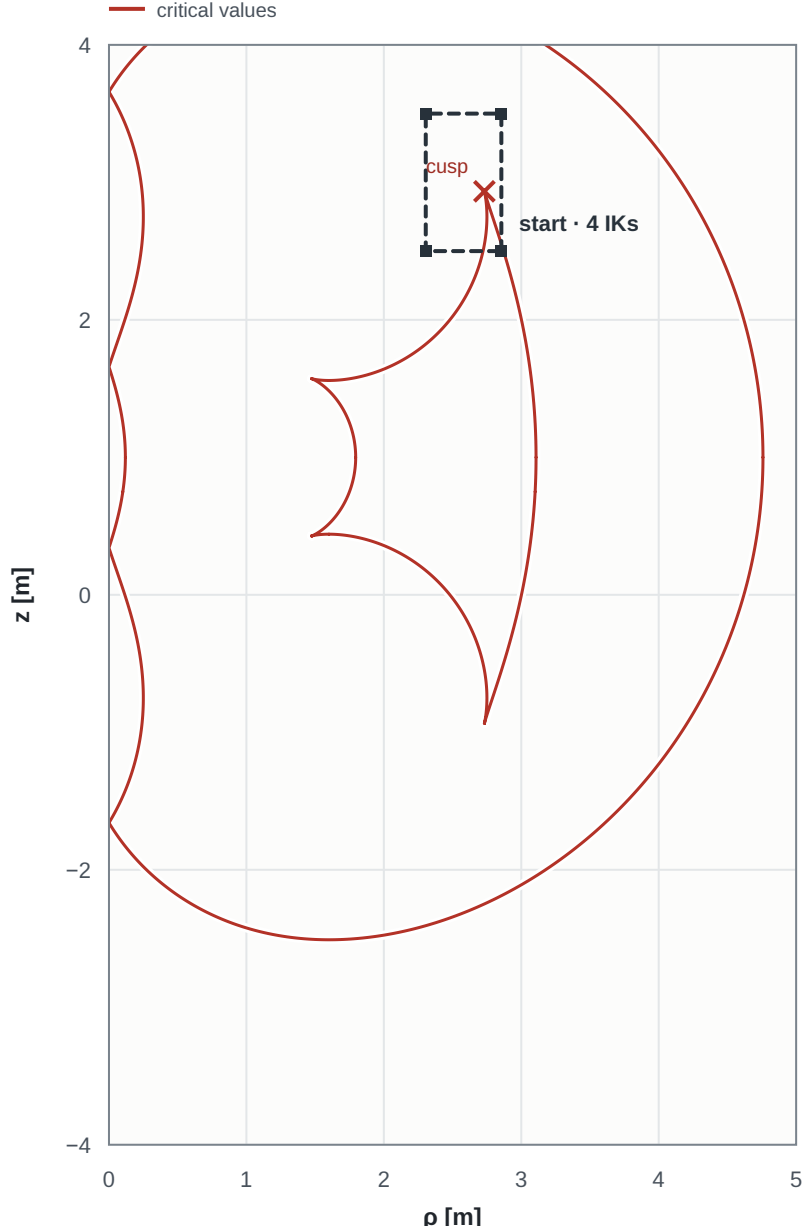
+

Reset axes

Fit path / cusp

Wheel: zoom · Shift-drag: pan

critical values



Finalize once: every following comparison uses this exact path.

center ρ

center z

width

height

rotation

2.58 m

3.00 m

1.00 m

0.55 m

90 °

Draw vertices

Undo vertex

Finalize shared path

Verified example

center (2.58, 3.00) m · 1.00 × 0.55 m · 90°

Start: 4 IKs · model cusp (2.7316, 2.9346) m

1 closed joint loop(s) · 2 fold termination(s) · 1 nonsingular change(s). Lift = a continuous joint trajectory that follows the task-space path.

Path feasibility is a property of the selected solution sheet

Regular

The selected joint path remains nonsingular and closes.



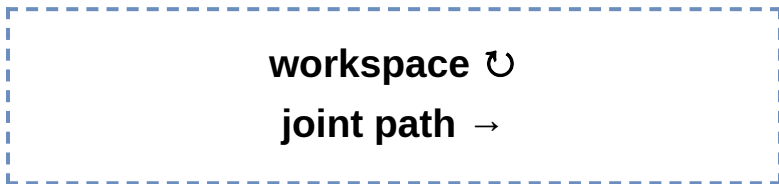
Branch-infeasible

The selected joint trajectory reaches a fold or another continuation barrier; a different starting IK may remain feasible.



Nonsingular change

The lift stays regular but returns to a different IK of the starting point.



The Cartesian command alone does not identify which of these joint-space outcomes the robot will follow.

A closed task-space loop can return to the same joint configuration

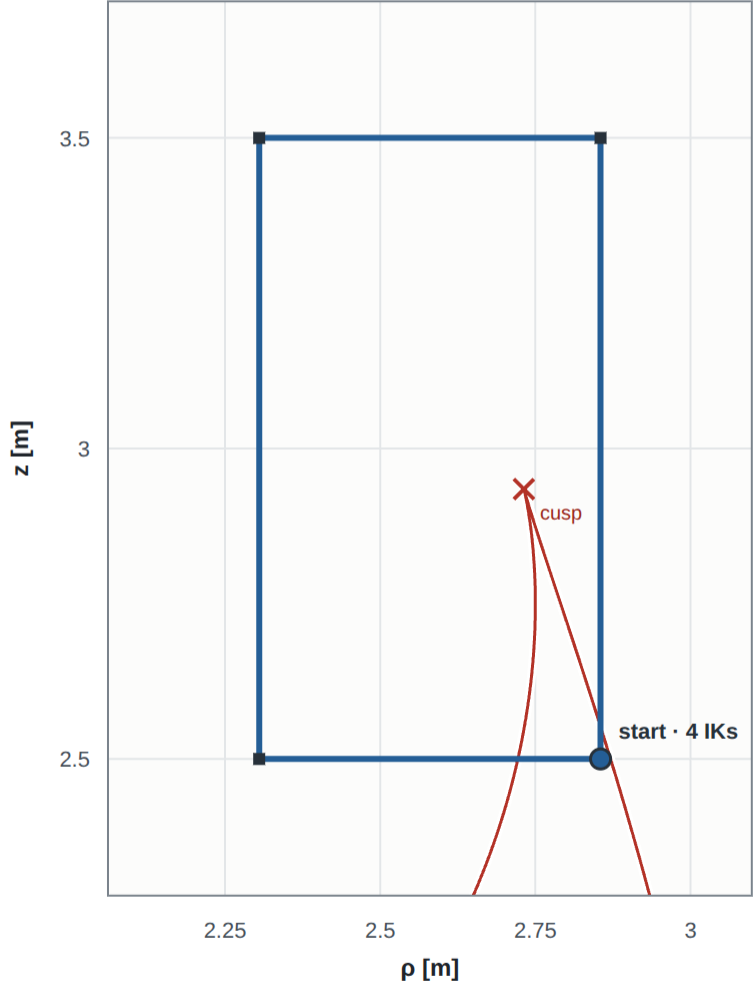
The continuation returns near the starting IK and has nonzero determinant at every sampled point; refine the sampling before treating the lift as validated.

Shared task path · $y = 0$

Zoomed

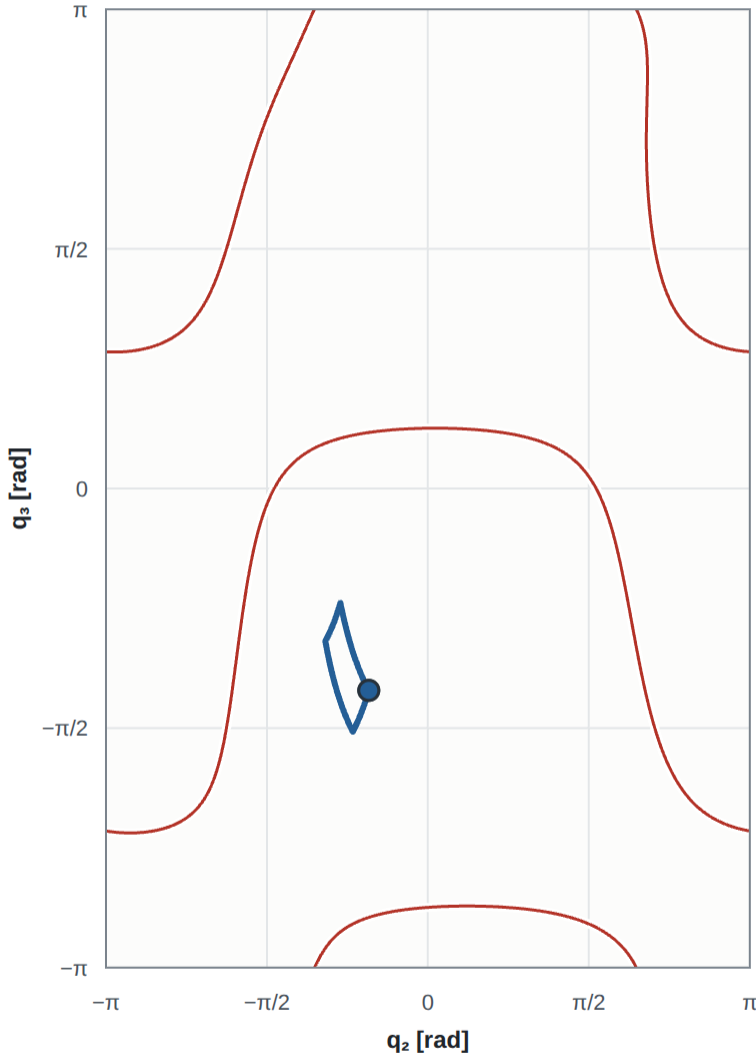
Full

critical values

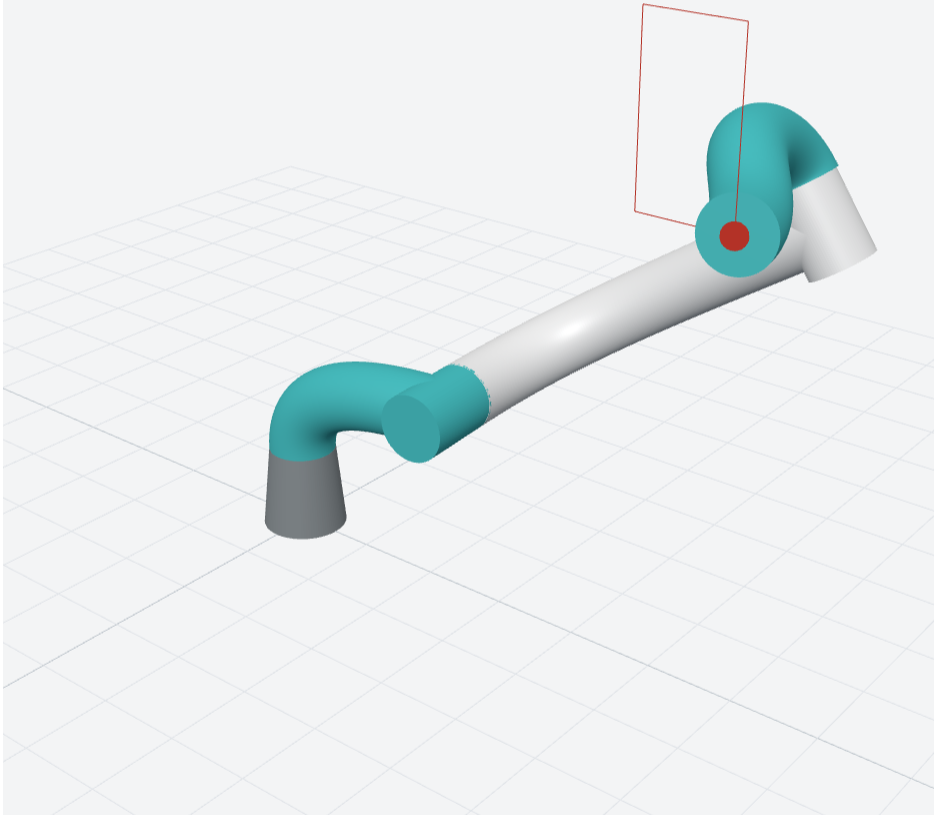


Lift = continuous joint trajectory

det $J_p = 0$



Custom 3R · exact native URDF geometry and STL visuals



STL opacity 100%

Start IK 4 · first traversal · closed joint loop

IK 4 · closed joint loop

Replay

Reset

closed joint loop · joint endpoint separation 0.0000 rad · min sampled $|\det J_p|$ 6.76e+0 m³. Adaptive continuation bounds joint increments below 0.055 rad.

sample 519/519

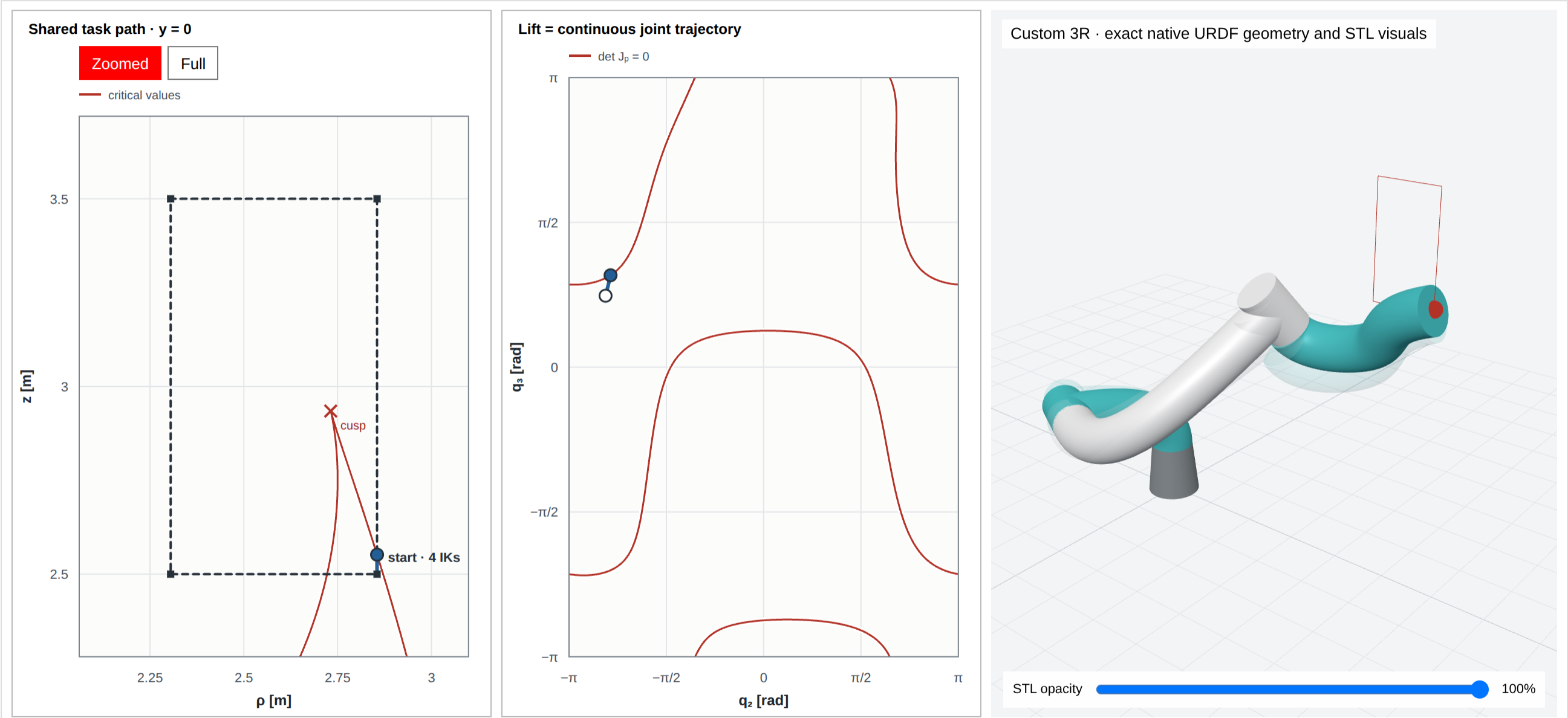
$q_1, q_2, q_3 = 4.1^\circ, -33.0^\circ, -75.8^\circ$

$|\det J_p| = 9.66413 \text{ m}^3$

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Track a starting IK whose continuation cannot complete the path

The workspace command continues, but this IK branch merges with another solution at $\det J_p = 0$.



Start IK 1 · first traversal · fold termination

IK 1 · fold termination

ReplayReset

fold termination · Selected IK reaches a singular fold; no regular continuation on this branch. · min sampled $|\det J_p|$ 1.71e-4 m³. Adaptive continuation bounds joint increments below 0.055 rad.

sample 17/17

$q_1, q_2, q_3 = -118.5^\circ, -141.6^\circ, 57.1^\circ$

$|\det J_p| = 0.00017 \text{ m}^3$

Compare a second starting IK on exactly the same task-space path

The measured endpoint and failure cause belong to this starting IK. Changing the master path can change which trajectories exist.

Shared task path · $y = 0$

Zoomed

Full

critical values

Lift = continuous joint trajectory

$\det J_p = 0$

Custom 3R · exact native URDF geometry and STL visuals

Start IK 2 · first traversal · fold termination

IK 2 · fold termination

Replay

Reset

fold termination · Selected IK reaches a singular fold; no regular continuation on this branch. · min sampled $|\det J_p|$ 1.71e-4 m³. Adaptive continuation bounds joint increments below 0.055 rad.

sample 17/17

$q_1, q_2, q_3 = -118.5^\circ, -141.6^\circ, 57.1^\circ$

$|\det J_p| = 0.00017 \text{ m}^3$

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One traversal changes IK solution without crossing a singularity

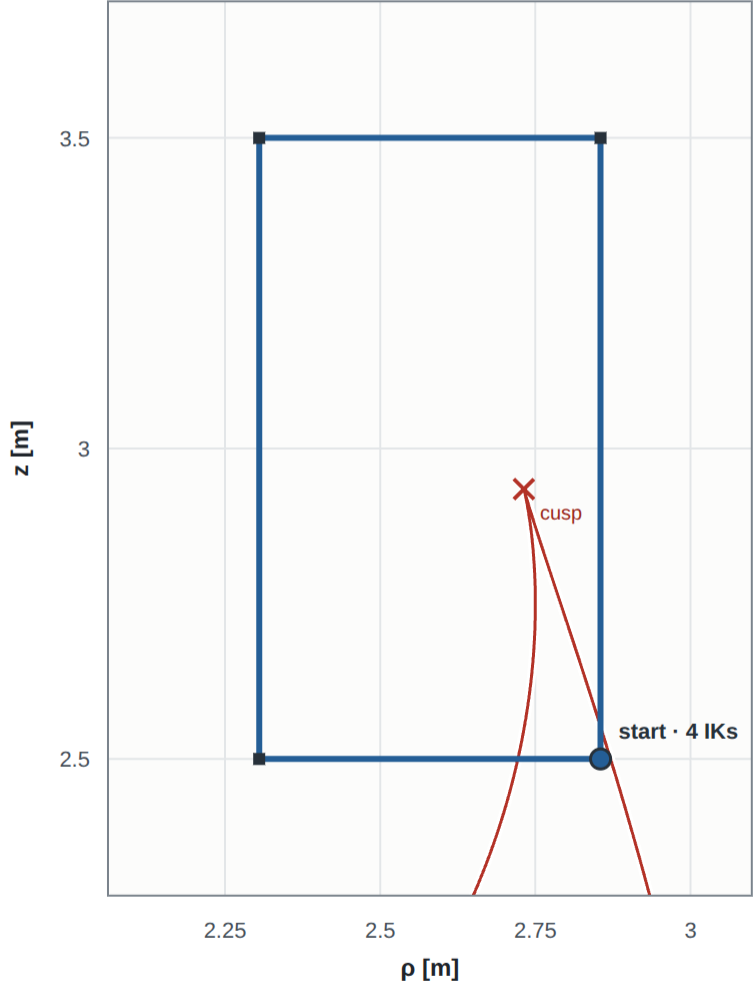
The workspace loop closes while its regular joint-space lift remains open.

Shared task path · $y = 0$

Zoomed

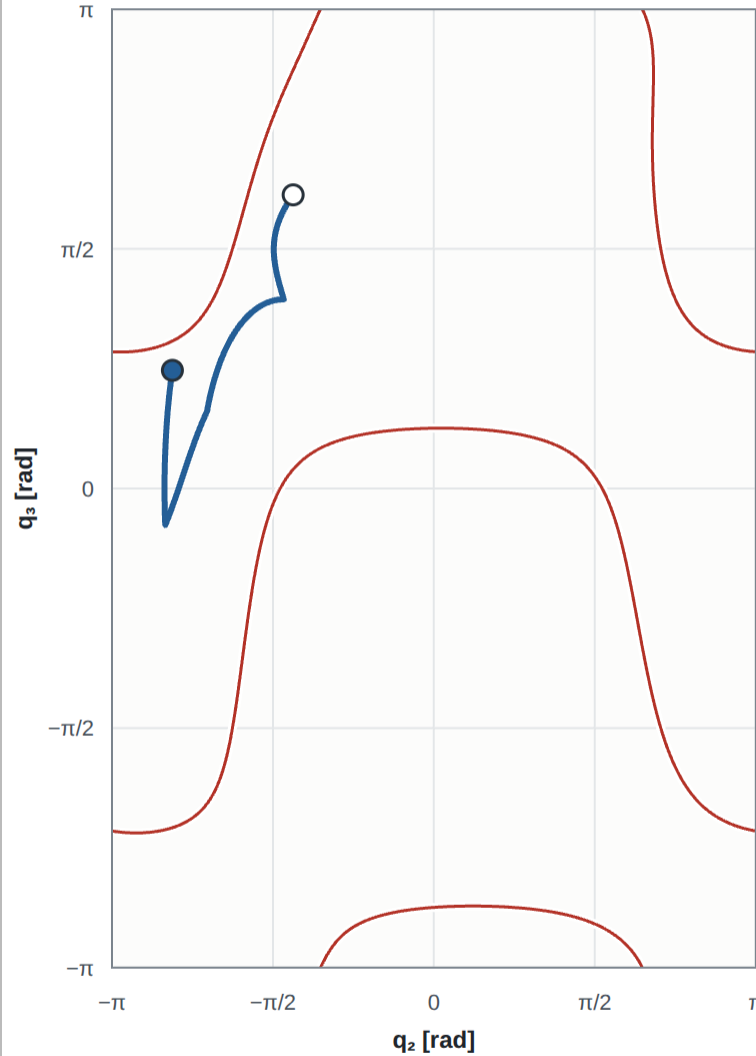
Full

critical values

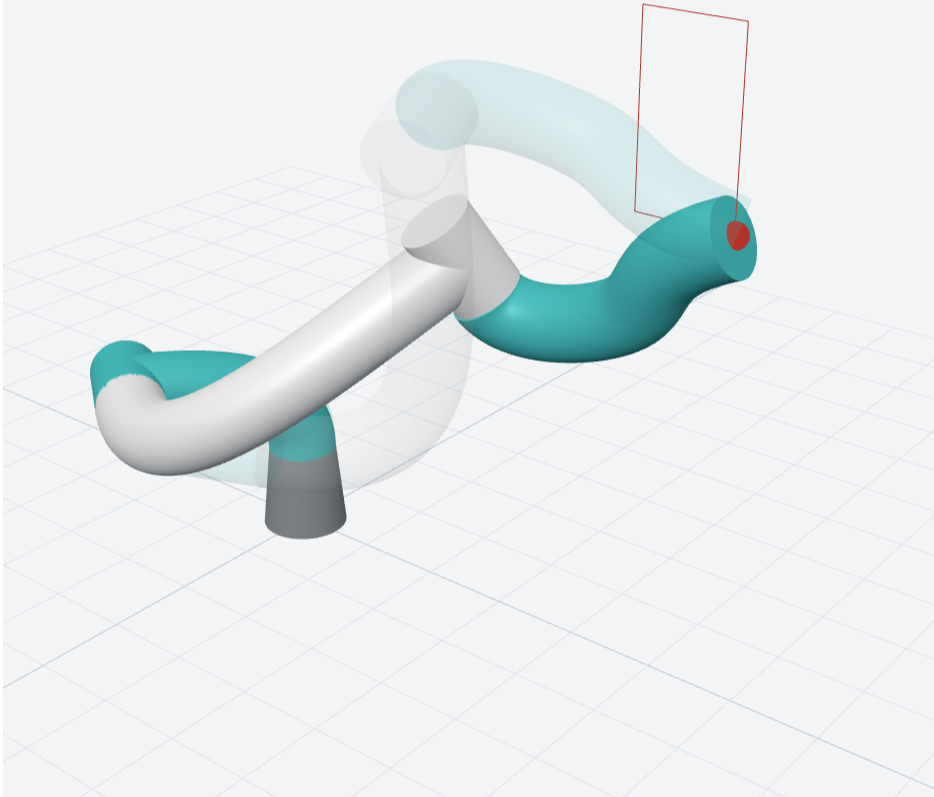


Lift = continuous joint trajectory

det $J_p = 0$



Custom 3R · exact native URDF geometry and STL visuals



STL opacity 100%

IK 3 · nonsingular change ▾

Replay

Reset

Start IK 3 · first traversal · nonsingular change

nonsingular change · joint endpoint separation 1.9316 rad · min sampled $|\det J_p|$ 1.18e+0 m³. Adaptive continuation bounds joint increments below 0.055 rad.

sample 519/519
 $q_1, q_2, q_3 = -126.4^\circ, -146.2^\circ, 44.4^\circ$
 $|\det J_p| = 1.18059 \text{ m}^3$

The second lap starts from the configuration reached after lap one

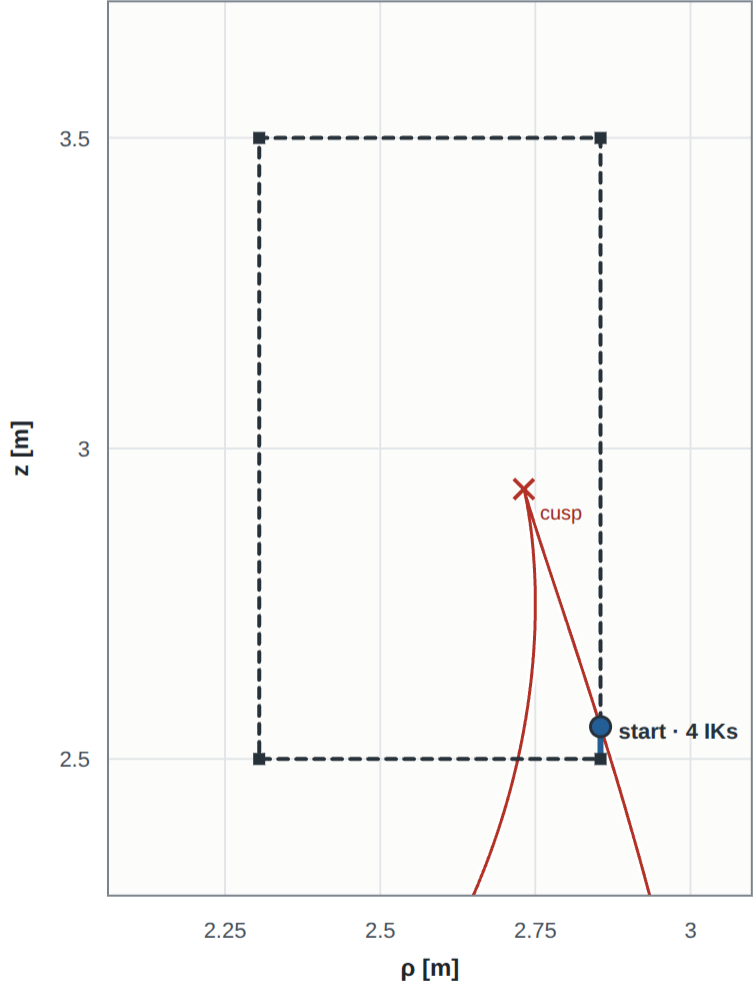
Play lap 1, then repeat the same path from its new endpoint. Test whether the second joint trajectory closes, changes solution again, or stops.

Shared task path · $y = 0$

Zoomed

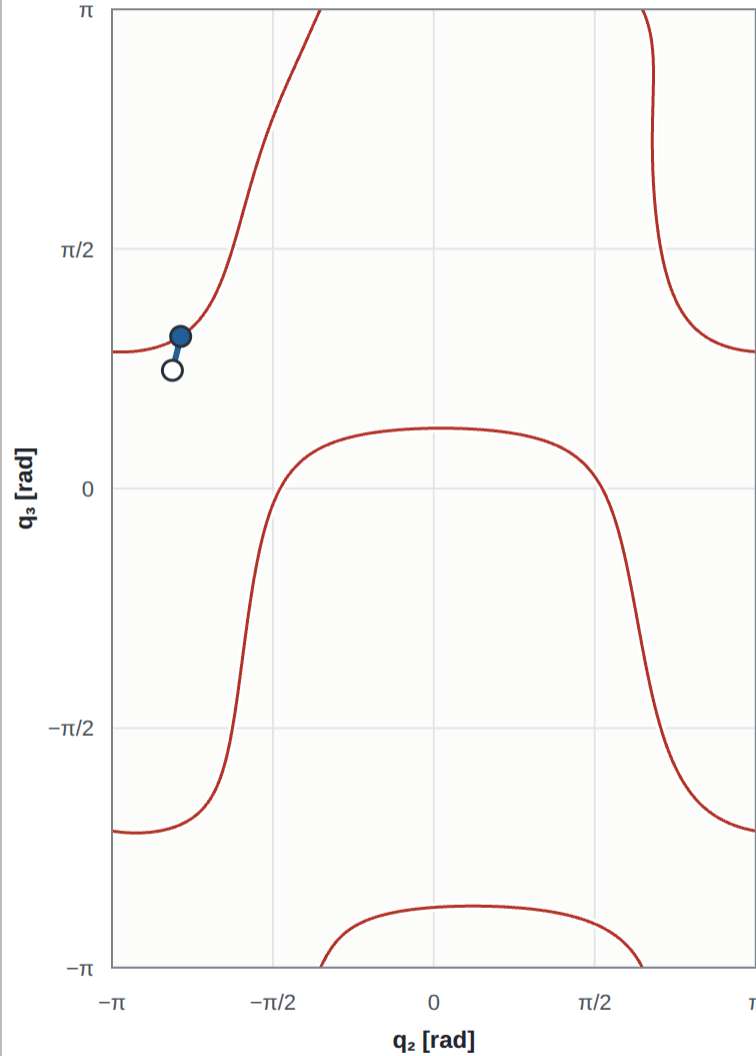
Full

critical values

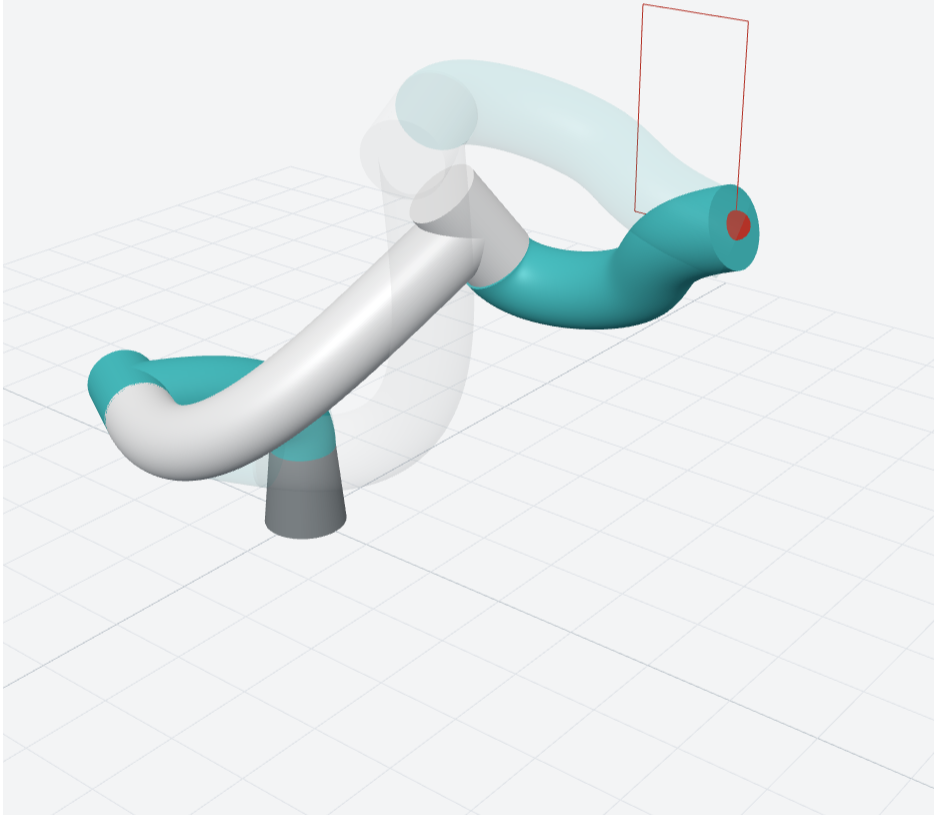


Lift = continuous joint trajectory

$\det J_p = 0$



Custom 3R · exact native URDF geometry and STL visuals



STL opacity 100%

IK 3 · nonsingular change ▼

Replay

Reset

Start IK 3 · second traversal · fold termination

Lap 2: fold termination · Selected IK reaches a singular fold; no regular continuation on this branch. · min sampled $|\det J_p|$ 1.71e-4 m³. Adaptive continuation bounds joint increments below 0.055 rad.

sample 17/17

$q_1, q_2, q_3 = -118.5^\circ, -141.6^\circ, 57.1^\circ$

$|\det J_p| = 0.00017 \text{ m}^3$

ABB GoFa CRB 15000: the wrist offset changes the inverse geometry



Six axes, offset wrist

The supplied CRB 15000-5/0.95 has a 0.080 m offset between its last wrist axes. The three wrist axes do not meet at one common centre.

Enumerate the configurations

The uploaded analytical IK evaluates a degree-16 polynomial, reconstructs the joint angles, and verifies each result by forward kinematics. We count real solutions and joint-limit-admissible solutions separately.

Track connectivity

Numerical labels such as “IK 1” are local to one pose. Follow the continuous joint trajectory to discover which solution it reaches at the end.

The joint-limit interval matters: $q_3 \in [-225^\circ, 85^\circ]$. A legal configuration below -180° must not be discarded merely because an equivalent angle was wrapped into another interval.

Count the inverse solutions in an x–y slice

The default map samples 100,000 poses at $z = 0.50$ m with fixed orientation $R_z(0)R_y(-0.28)R_x(1.47)$ (radians). It contains regions with 4, 6, 8, 10, 12, 14 and 16 real IKs before joint limits are applied.

Explore both scales: Central detail loads a saved 100,000-pose map for the default slice to resolve the 14- and 16-IK regions. Other heights and orientations are calculated on demand. Click a point to enumerate its solutions; compare the counts before and after joint limits.

Tool position · x-y slice · z = 0.5 m

Full slice

Central detail

☒ q₄ limit boundary

Dashed red: q₄ = ±180° for at least one legal IK. Other IKs can remain feasible across this boundary.

ABB CRB 15000 · white/grey CAD · selected IK solid

☒ Labels

STL opacity 100%

z [m]0.5Yaw [°]0

Pitch [°]-16.042818Roll [°]84.224796

MapDetailed · 100,000 posesCalculate slice

Verified default slice

CountAll real IKsShow all IKs

Selected only

8 IKs

10 IKs

12 IKs

14 IKs

16 IKs

Select point

Draw path

Example loop

Close loop

Undo vertex

Clear path

Track selected IK

Compare all starts

Central detail loaded · 100,000 saved poses. 16 verified IKs · 10 within joint limits.

► Joint limits and branch continuity

► Orientation and sampling

☒ IK 1

☒ IK 2

☒ IK 3 · outside limits

☒ IK 4

☒ IK 5 · outside limits

☒ IK 6

☒ IK 7

☒ IK 8

☒ IK 9

☒ IK 10 · outside limits

☒ IK 11

☒ IK 12 · outside limits

☒ IK 13

☒ IK 14 · outside limits

☒ IK 15 · outside limits

☒ IK 16

Play

Reset

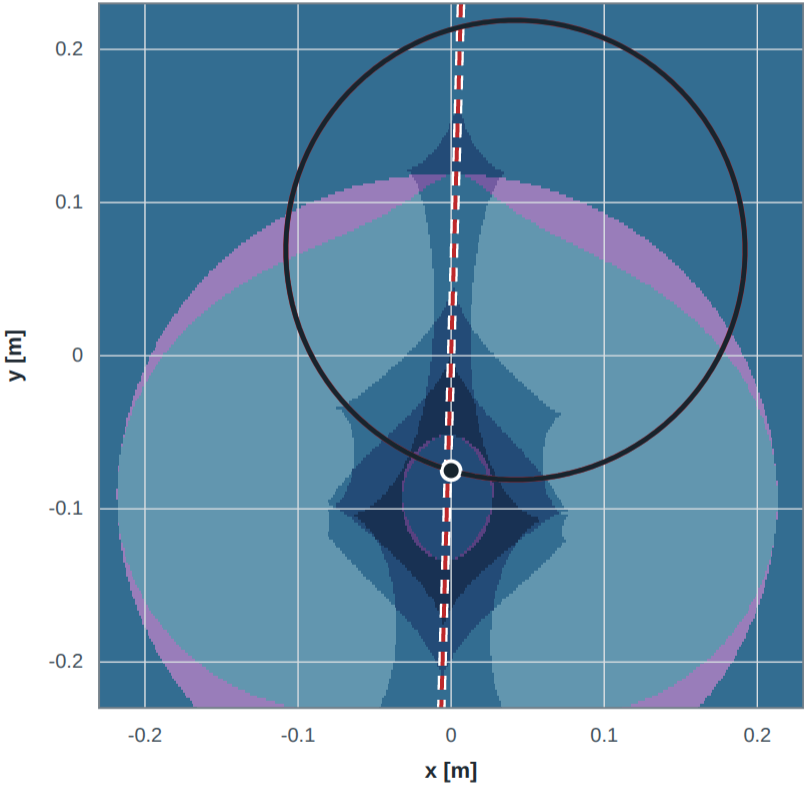
Draw one task path; compare the joint trajectories from its IKs

Choose the tool orientation and height z , then calculate the x–y map. The selected point and drawn path carry forward from the previous map. Each path holds the chosen orientation and height constant.

Compare the starts: at the default height and orientation, the 30 cm example loop crosses a region with eight IKs inside joint limits. Follow each starting IK: the map count alone cannot certify a continuous legal path. Inspect why each continuation stops.

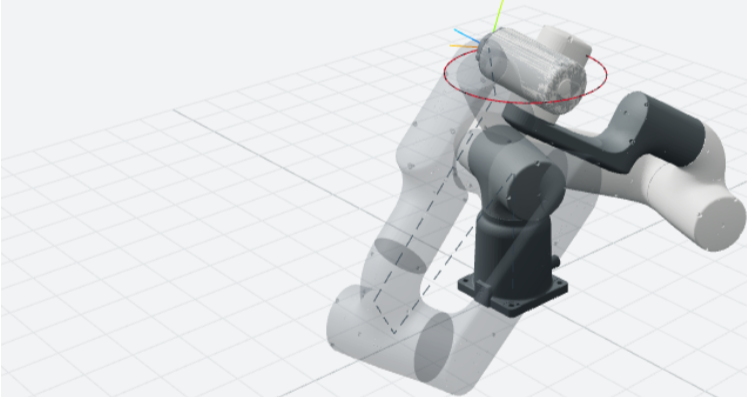
Tool position · x-y slice · $z = 0.5$ m

Full sliceCentral detail☒ q_4 limit boundary



Dashed red: $q_4 = \pm 180^\circ$ for at least one legal IK. Other IKs can remain feasible across this boundary.

ABB CRB 15000 · white/grey CAD · selected IK solid



☒ Labels

STL opacity 100%

z [m]0.5

Yaw [°]0

Pitch [°]-16.042818

Roll [°]84.224796

MapDetailed · 100,000 posesCalculate slice

Verified default slice

CountInside joint limitsShow all IKs

Selected only

6 IKs

7 IKs

8 IKs

9 IKs

10 IKs

11 IKs

12 IKs

Select point

Draw path

Example loop

Close loop

Undo vertex

Clear path

Track selected IK

Compare all starts

Central detail loaded · 100,000 saved poses. 3/16 starting IKs complete the path. Select a result to animate it. Selected IK 1: The selected branch follows the full path within the tested joint-limit and nonsingularity margins.

► Joint limits and branch continuity

► Orientation and sampling

IK 1: complete

☒ IK 1

☒ IK 2

☒ IK 3 · outside limits

☒ IK 4

☒ IK 5 · outside limits

☒ IK 6

☒ IK 7

☒ IK 8

☒ IK 9

☒ IK 10 · outside limits

☒ IK 11

☒ IK 12 · outside limits

☒ IK 13

☒ IK 14 · outside limits

☒ IK 15 · outside limits

☒ IK 16

Replay

Reset

361/361 · s = 1.000

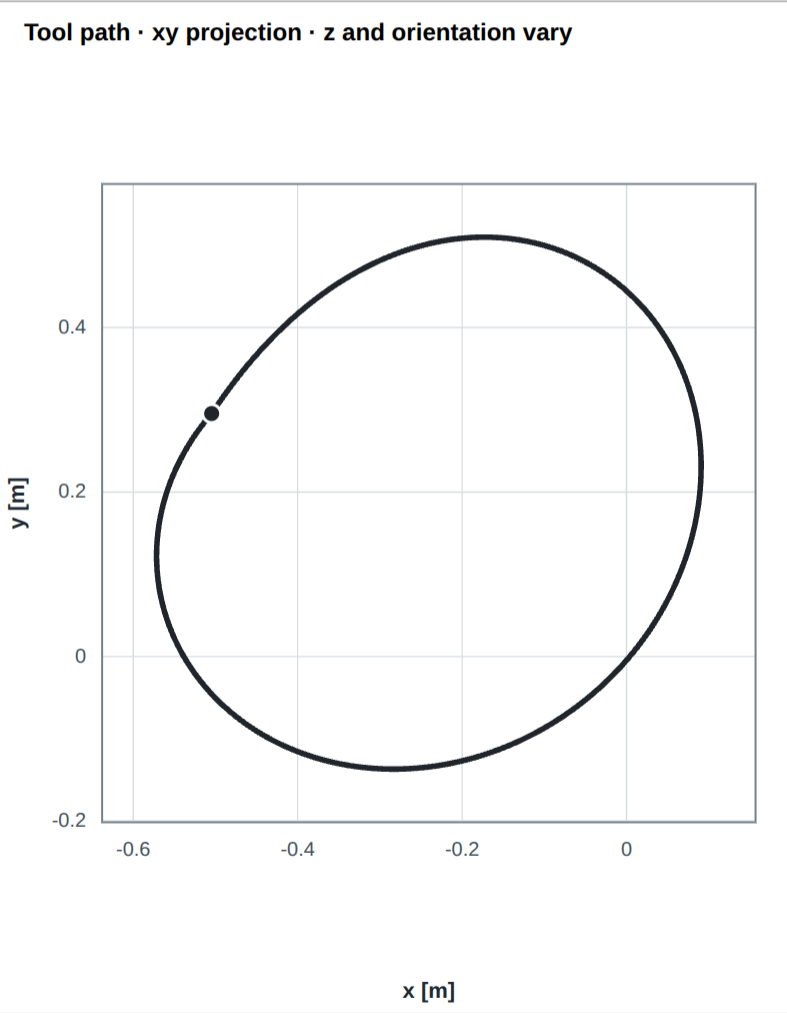
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Same final tool pose, a different joint configuration, no joint-limit crossing

A searched connection joins two IKs of the same full pose. This demonstration allows orientation to vary along the task-space loop. The transparent robot stays at the starting configuration.

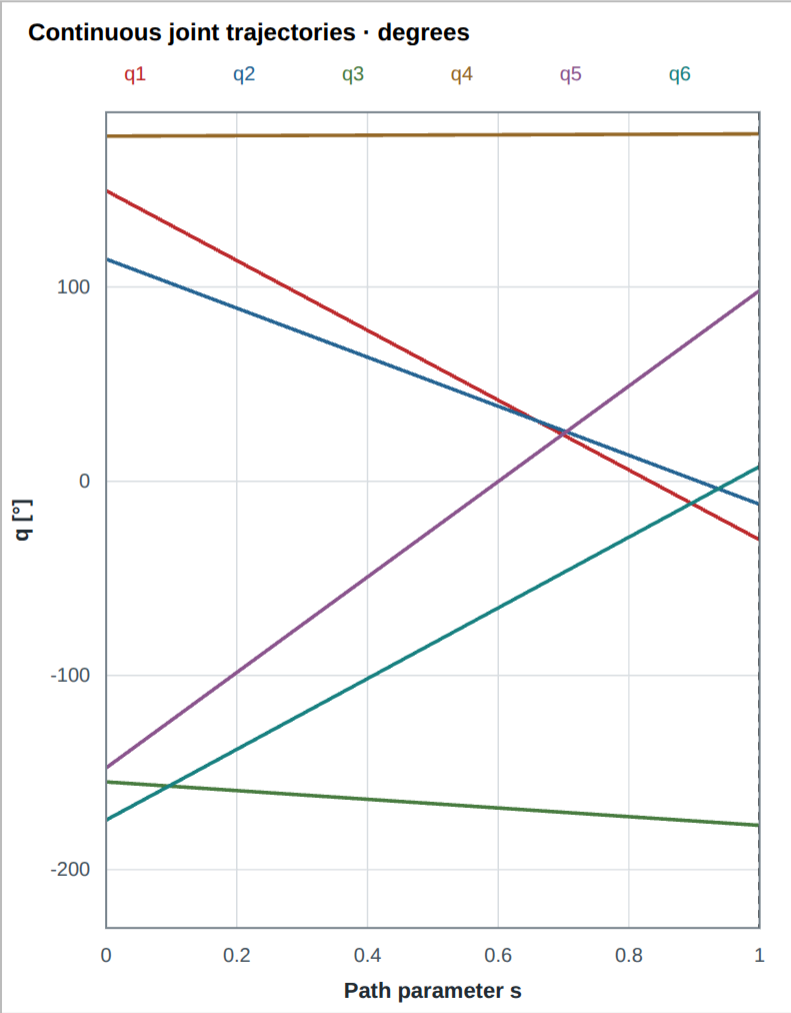
Search: enumerate IK pairs of the same pose, connect legal joint configurations, and map the connection through FK. Here a determinant derivative bound verifies the entire joint segment; all 401 animation poses are rechecked by IK.

Tool path · xy projection · z and orientation vary



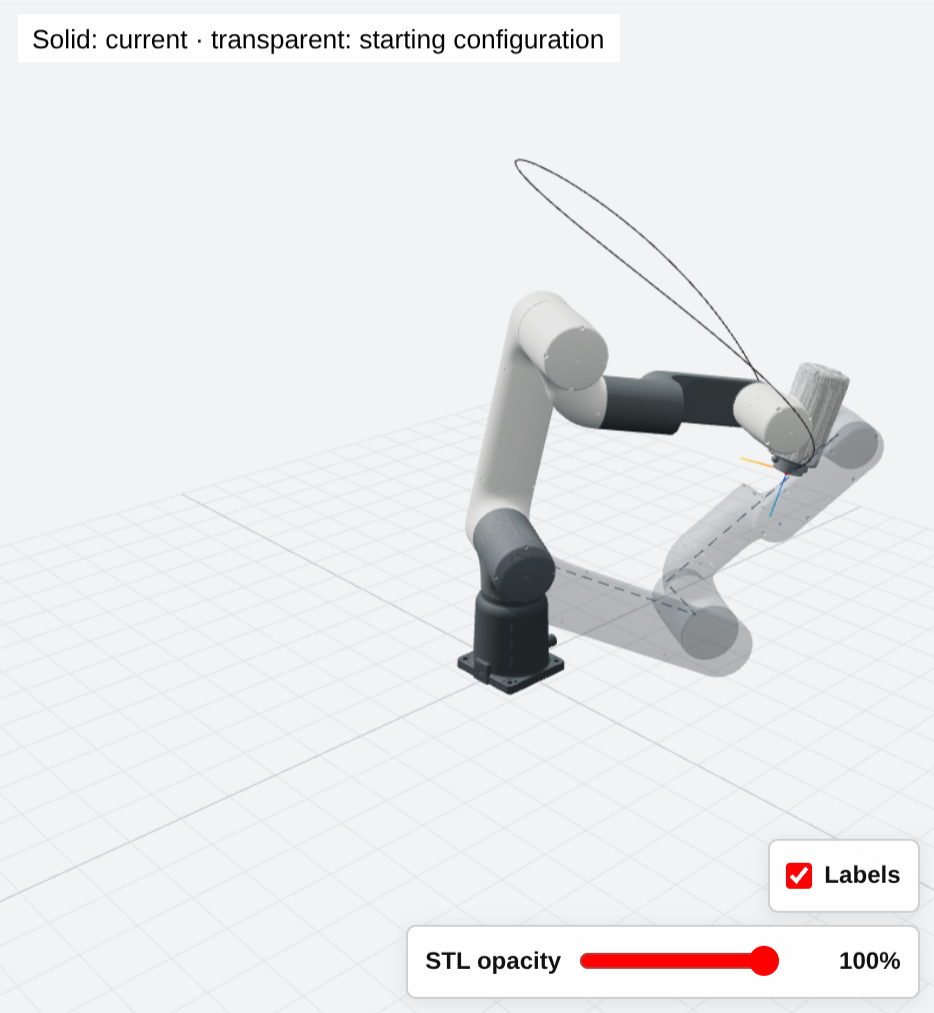
A 2D plot showing the xy projection of the tool path. The x-axis ranges from -0.6 to 0 m, and the y-axis ranges from -0.2 to 0.4 m. The path is a closed loop, with a single point highlighted on the left side.

Continuous joint trajectories · degrees



A line graph showing the continuous joint trajectories for joints q1 through q6 over a path parameter s from 0 to 1. The y-axis represents joint angles q in degrees, ranging from -200 to 100. q1 (red) decreases from ~150 to ~-50. q2 (blue) decreases from ~110 to ~-10. q3 (green) decreases from ~-150 to ~-170. q4 (orange) is constant at ~178. q5 (purple) increases from ~-170 to ~100. q6 (teal) increases from ~-170 to ~10.

Solid: current · transparent: starting configuration



A 3D visualization of a robotic arm. The current configuration is shown in solid grey, and the starting configuration is shown in transparent grey. The arm is positioned on a grid floor. A legend indicates 'Solid: current · transparent: starting configuration'. A 'Labels' checkbox is checked. An 'STL opacity' slider is set to 100%.

Replay

Reset

Show endpoint

IK 7 → IK 5 · 401/401 samples · joint separation 6.584 rad · pose closure 1.1e-15 m / 1.2e-15 rad · whole-path |det J| > 0.00755 m³ · min limit margin 1.17°.

Start q [°]: 149.6, 114.4, -154.8, 177.7, -147.6, -174.4

Now q [°]: -30.1, -11.9, -177.2, 178.8, 98.2, 7.6

End q [°]: -30.1, -11.9, -177.2, 178.8, 98.2, 7.6

One robot exhibits several distinct planning outcomes

MoveL means “move linearly”: the tool centre point follows a straight Cartesian segment, $\mathbf{p}(s) = (1 - s)\mathbf{p}_A + s\mathbf{p}_B$, with $0 \leq s \leq 1$. The tool orientation interpolates between its endpoint orientations; equal endpoint orientations keep it fixed.

Example in this lecture	Verified outcome	Planning lesson
Fixed-orientation x–y loop	At $z = 0.50$ m, the default loop starts with 16 real IKs, including 10 inside the joint limits. Two starting IKs complete the loop.	Pose-wise IK counts do not establish the feasibility of a continuous joint path.
Legal nonsingular connection	Same full endpoint pose; different joint configurations; positive determinant and limit margins.	Task-space closure does not imply joint-space closure. Repeatability requires its own check.
MoveL with five legal continuations	Eight initial IKs and ten final IKs. Five initial solutions complete the line inside the limits.	Enumerate and compare whole trajectories before choosing the initial IK.
MoveL with no legal continuation	Eight legal starts, but none can finish the prescribed line within the joint limits.	Changing the starting IK is insufficient. Change the path, orientation, or task placement.

Reachable endpoints do not guarantee a feasible path between them. Compare continuous joint trajectories over the whole command: an IK may close, change solution, meet a singularity, or reach a joint limit.

Reference: Elias & Wen (2025), *Path Planning and Optimization for Cuspidal 6R Manipulators*.

Turn inverse solutions along the path into a connectivity graph

Nodes · all IKs at each sample

$$\mathcal{Q}_k = \{\mathbf{q}_k^{(1)}, \mathbf{q}_k^{(2)}, \dots\}$$

Each task pose is a layer. Retain its real inverse solutions and mark those outside joint limits.

Edges · verified continuations

Connect adjacent layers only when the joint motion is continuous and satisfies the required limits and singularity margin. Equal determinant sign alone does not establish connectivity.

Search · an entire joint path

Search from the first layer to the last. If no chain is found, refine uncertain connections before declaring the command infeasible. For a closed task loop, compare final and initial configurations and test another lap.

For a cuspidal 6R robot, local S/E/W-style labels do not provide a complete global description. Check the actual joint connections. Regular closed loops, solution-changing loops, and failed continuations can occur on the same robot.

Global analysis connects solutions before choosing the joint trajectory

- 1 Receive the workspace path or trajectory from a precomputed command or a real-time policy.
- 2 Enumerate candidate IKs and connect continuous joint trajectories over the available path or planning horizon.
- 3 Track branch identity, singularity distance, joint limits, and collisions.
- 4 For closed task loops, test joint-space closure and repeatability explicitly.

Hybrid conclusion: enumeration supplies the candidate IKs; continuation connects them locally; comparing full trajectories reveals limits, singularities, endpoint changes, and repeatability. For a known path, global analysis can choose a viable start or identify an infeasible command. With a real-time policy, feasibility must be reassessed as new commands arrive. The demonstrations check kinematics and joint limits; collision and dynamics checks are additional planning stages. Lecture 08 adds redundancy.